

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:20

Annexure A
SHORT ANSWER TEST

Criteria	Conceptual Understanding(3)	Accuracy of Answer(2.5)	Application of Knowledge(2)	Completeness of Response(1.5)	Clarity & Presentation(1)	Total(10)
Weightage	30%	25%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						

Code of Conduct:

I S.SHAMAH (192573002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

shamah
Signature of the student:

Annexure A
Short Answer Test

1. Suppose that a data warehouse consists of the three dimensions time, doctor, and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit.

(a) Enumerate three classes of schemas that are popularly used for modeling data warehouses.

(b) Draw a schema diagram for the above data warehouse using one of the schema.

(c) To obtain the same list, write an SQL query assuming the data is stored in a relational database with the schema fee (day, month, year, doctor, hospital, patient, count, charge).

(5 Marks)

2. Suppose that a data warehouse for Big University consists of the following four dimensions: student, course, semester, and instructor, and two measures count and avg grade. When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the avg grade measure stores the actual course grade of the student. At higher conceptual levels, avg grade stores the average grade for the given combination. (5 Marks)

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [student, course, semester, instructor], what specific OLAP operations

(e.g., roll-up from semester to year) should one perform in order to list the average grade of CS courses

for each Big University student.

(c) If each dimension has five levels (including all), such as “student < major < status < university < all”, how many cuboids will this cube contain (including the base and apex cuboids)?

3. Design a data warehouse for a regional weather bureau. The weather bureau has about 1,000 probes, which are scattered throughout various land and ocean locations in the region to collect basic weather data, including air pressure, temperature, and precipitation at each hour. All data are sent to the central station, which has collected such data for over 10 years. Your design should facilitate efficient querying and on-line analytical processing, and derive general weather patterns in multidimensional space. (5 Marks)

4. A data cube, C , has n dimensions, and each dimension has exactly p distinct values in the base cuboid. (5 Marks)

Assume that there are no concept hierarchies associated with the dimensions.

(a) What is the maximum number of cells possible in the base cuboid?

(b) What is the minimum number of cells possible in the base cuboid?

(c) What is the maximum number of cells possible (including both base cells and aggregate cells) in the

data cube, C ?

(d) What is the minimum number of cells possible in the data cube, C ?

- 1) Suppose that a data warehouse consists of the three dimensions time, doctor and patient and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit.
- a) Enumerate three classes of schemas that are popularly used for modeling data warehouses.
- b) Draw a schema diagram for the above data warehouse using one of the schema.
- c) To obtain the same list, write an SQL query assuming the data is stored in a relational database with the schema fee.

Ans: DATA WAREHOUSE: TIME, DOCTOR, PATIENT

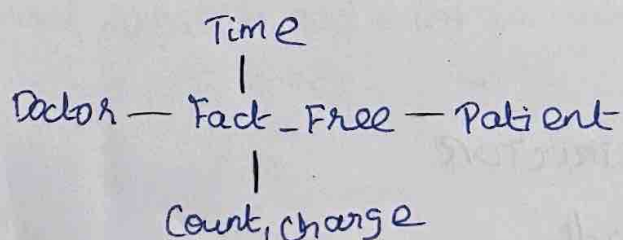
- a) Three Popular schemas:

* Star schema

* Snowflake Schema

* Fact Constellation Schema.

- b) Star schema:



c) SQL Query:

```
SELECT day, month, year, doctor, hospital, patient, count, charge  
FROM Fee;
```

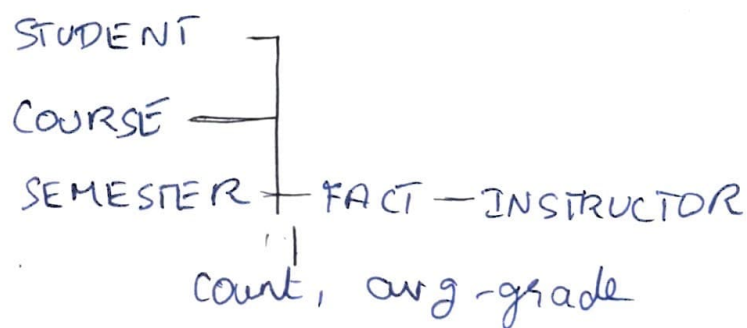
2) Suppose that a data warehouse for Big University consists of the following four dimensions: student, course, semester and instructor, and two measures count and avg grade, when at the lowest conceptual level. At higher conceptual levels, avg grade stores the average grade for the combination.

a) Draw a snowflake schema diagram for the data warehouse

b) Starting with the base cuboid, what specific OLAP operation.

c) If each dimension has five levels (including all), such as "student < major < status < university < all", how many cuboids will this cube contain.

a) Snowflake Schema:



b) OLAP operations:

Starting from:

[student, course, semester, instructor]

Perform:

- i) Roll-up: semester $\rightarrow$ year
- ii) Roll-up: instructor $\rightarrow$ all
- iii) Roll-up: course $\rightarrow$ department
- iv) slice/Dice: select CS courses
- v) Display student and average grade.

c) Number of cuboids:

Each dimension has 5 levels:

$$5^4 = 625 \text{ cuboids.}$$

3) Design a data warehouse for a regional weather bureau.

All data are sent to the central station, which has collected such data for over 10 years. Your design should facilitate efficient querying and on line analytical processing and derive general weather patterns in multidimensional space.

Ans: Weather Bureau Data Warehouse

Dimensions:

- * Time : Hour $\rightarrow$ Day $\rightarrow$ Month $\rightarrow$ Year $\rightarrow$ All
- * Location : Probe $\rightarrow$ Area $\rightarrow$ Region $\rightarrow$ All
- * Weather Type : Air Pressure, Temperature, Precipitation.

Fact Table.

WEATHER - FACT

Time - id

Location - id

Pressure

Temperature

Precipitation

Advantages:

- * Stores 10+ years of historical data
- * Supports fast OLAP queries.
- * Enable analysis of weather patterns.
- * Supports roll up from hour $\rightarrow$ day $\rightarrow$ month $\rightarrow$ year
- * Supports analysis by probe, area and region.

A data cube, C has n dimensions and each dimension has exactly P distinct values in the base cuboid.

- a) what is the maximum number of cells possible in the base cuboid?
- b) what is the minimum number of cells possible in the base cuboid?
- c) what is the maximum number of cells possible.
- d) what is the minimum number of cells possible in the data cube, C ?

Ans:

- a) Maximum number of base cells - P^n .
- b) Minimum number of base cells - P .
- c) Maximum total cells in the cube - $(P+1)^n$.
- d) Minimum total cells in the cube.

Minimum base cells = P , and all aggregate combinations are possible:

$$P + n.$$